Bounds recap

1. Markov Inequality

$$\forall x > 0, \quad P[x \geq a] \leq \frac{E[x]}{a}$$

2. Chebyshev Inequality

$$P[|x - E[x]| \geq a] \leq \frac{\sigma_x^2}{a^2}$$

3. Hoeffding's Inequality

$\forall \{x_i\}: x_i \in [a_i, b_i]$, $\{x_i\}$ are independent

$$P\left[\left|\frac{1}{N}\sum_{i=1}^{N}(x_i - E[x_i])\right| > a\right] \leq 2e^{\frac{-2a^2N^2}{\sum_{i=1}^{N}(b_i - a_i)}}$$

3 bis. If $\{x_i\}$ are iid with $\mu \triangleq E[x_i]$

$$P[|\bar{x} - \mu| > a] \leq 2e^{-\frac{Na^2}{\Delta^2}}, \quad \Delta = b_i - a_i$$

3 ter $\{x_i\}$ iid $\Delta = 1$

$$P[|\bar{x} - \mu| > a] = 2e^{-a^2N^2}$$

4. Chernoff bound

$\bar{x}_N \triangleq \frac{1}{N}\sum_{i=1}^{N} x_i$ with $\{x_i\}$ iid rv

$$P[\bar{x}_N > a] \leq e^{-N \sup\{a\theta - \log M(\theta)\}}, \text{ where } \theta \in \mathbb{R}^+ \text{ and } M(\theta) = \text{Moment Generating function of } x_i \triangleq E[e^{\theta x_i}]$$

$$M(\theta) = E[e^{\theta x_i}], \quad M(0) = 1$$

$$M'(\theta) = \frac{dM(\theta)}{d\theta} = E\left[\frac{d}{d\theta}e^{\theta x_i}\right] = E[x_i e^{\theta x_i}]$$

$$M'(\theta) = E[x_i e^{0 \cdot x_i}] = E[x_i] \rightarrow 1^{st} \text{ order moment}$$

$$M''(\theta) \triangleq \frac{d}{d\theta}M'(\theta) = E[x_i^2 e^{\theta x_i}]$$

$$M''(0) = E[x_i^2] \rightarrow 2^{nd} \text{ order moment}$$

$$\bar{x}_N = \frac{1}{N}\sum_{i=1}^{N} x_i$$

$$P[\bar{x}_N \geq a] = P[\theta \textstyle\sum_{i=1}^{N} x_i \geq Na\theta] \; \because \theta > 0 \; ?$$

$$= P\left[e^{\theta \sum_{i=1}^{N} x_i} \geq e^{Na\theta}\right] \leq \frac{E\left[e^{\sum_{i=1}^{N} x_i}\right]}{e^{Na}} \quad \because \text{Markov inequality}$$

$$\frac{E\left[e^{\sum_{i=1}^{N} x_i}\right]}{e^{Na}} = \frac{E\left[\prod_{i=1}^{N} e^{x_i\theta}\right]}{e^{aN\theta}}$$

$$E\left[\prod_{i=1}^{N} e^{x_i\theta}\right] = e^{-aN\theta} \prod_{i=1}^{N} E[e^{\theta x_i}] \quad \because \{x_i\} \text{ are independent}$$

$$= e^{-aN\theta} \prod_{i=1}^{N} M_{x_i}(\theta)$$

If $\{x_i\}$ are i.d. $\Rightarrow M_{x_i}(\theta) = M(\theta)$, $\forall i = 1, 2, \ldots, N$

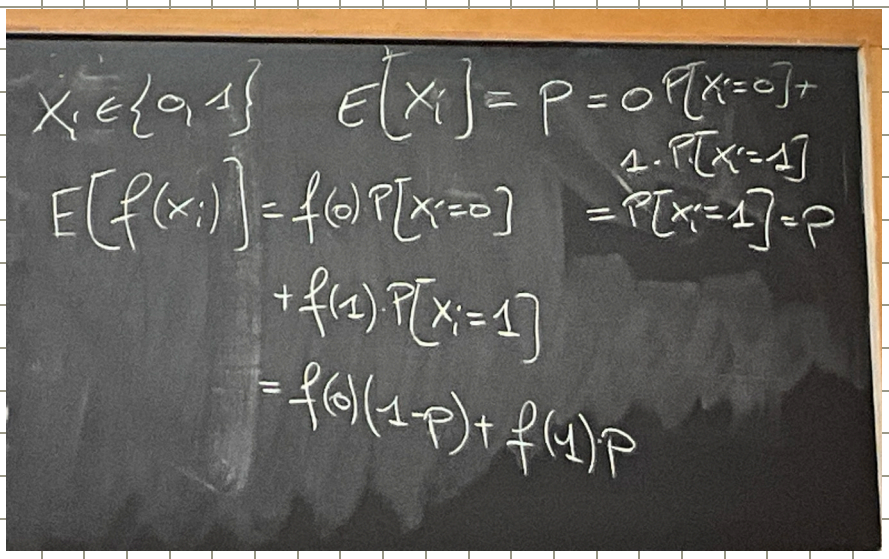
$$= e^{-aN\theta} M(\theta)^N = e^{-aN\theta + \log(M(\theta)^N)}$$

$$= e^{-aN\theta + N\log(M(\theta))} \quad \because \log \alpha^{\beta} = \beta \log \alpha = e^{-N(a\theta - \log M(\theta))}$$

$$P\left[\frac{1}{N}\sum_{i=1}^{N} x_i > a\right] \leq \underbrace{e^{-N(a\theta - \log M(\theta))}}$$

The strictest bound is obtained for θ^* s.t. $\theta^* = \arg\max_{\theta>0} \{a\theta - \log M(\theta)\}$

$$P[\ ] \leq e^{-N \sup_{\theta>0} \{a\theta - \log M(\theta)\}} \quad \leftarrow \text{very powerful bound}$$



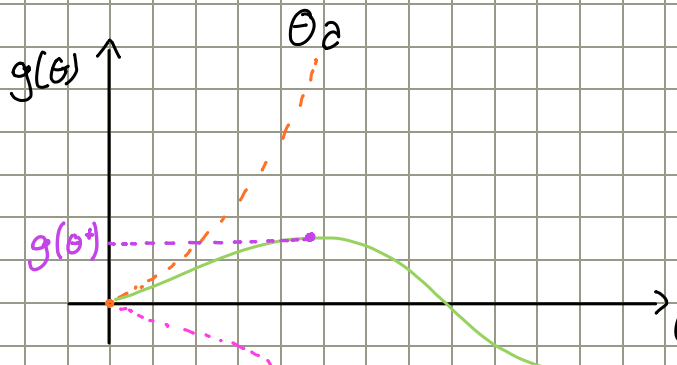
Special case: $X_i \in \{0, 1\}$

$P[x_i = 1] = p$; $P[x_i = 0] = 1-p \triangleq q$

$$M_{x_i}(\theta) = E[e^{\theta x_i}] = \underbrace{e^{\theta \cdot 0}}_{1} \cdot (1-p) + e^{\theta \cdot 1} \cdot p$$

$$= pe^{\theta} + 1 - p$$

$$P\left[\frac{1}{N}\sum_{i=0}^{N} x_i > a\right] \leq e^{-N \sup(\underbrace{\theta a - \log(pe^{\theta} + (1-p)}_{\triangleq g(\theta)})}$$



$$\theta \gg \log(pe^{\theta} + 1 - p)$$

$$\simeq \log(pe^{\theta}) = \log p + \underbrace{\log e^{\theta}}_{\theta}$$

$$\simeq \theta$$

It has a max somewhere:

$\frac{dg(\theta)}{d\theta} \stackrel{!}{=} 0$! = imposing this condition

$$a - \frac{pe^{\theta}}{pe^{\theta} + 1 - p} \stackrel{!}{=} 0 \Rightarrow ap^{\theta}e + a(1-p) - pe^{\theta} = 0$$

$$pe^{\theta} = \frac{a(1-p)}{1-a} \qquad \theta^* = \log \frac{a(1-p)}{p(1-a)}$$

$$g(\theta^*) = \theta^* a - \log(pe^{\theta} + 1 - p) = a \log \frac{a(1-p)}{p(1-a)} - \log\Big(\underbrace{\frac{a(1-p)}{1-a} + 1 - p}_{\frac{a(1-p)+(1-p)(1-a)}{1-a}}\Big)$$

$$g(\theta) = a \log\left(\frac{a(1-p)}{p(1-a)}\right) - \log\left(\frac{1-p}{1-a}\right)$$

$$= a \log \frac{a}{p} + \underbrace{a \log \frac{1-p}{1-a}} - \underbrace{\log \frac{1-p}{1-a}}$$

$$g(\theta^*) = a \log \frac{a}{p} + (a-1) \log \frac{1-p}{1-a} = a \log \frac{a}{p} + (1-a) \log \frac{1-a}{1-p}$$

$= \underset{\to D(a|p)}{\text{Kullback Leibler}}$ Divergence (KL-DV) (never heard of it)

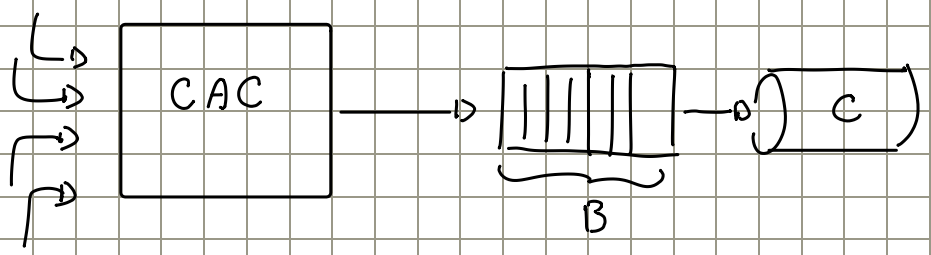
$x_i \in \{0, 1\} \qquad P[x_i = 1] = p$

$$P\left[\frac{1}{N} \sum x_i > a\right] \leq e^{-N \cdot D(a|p)}$$

$$D(f_x() \mid f_y()) \triangleq \int_{-\infty}^{+\infty} f_x(a) \log \frac{f_x(a)}{f_y(a)} \, da$$

These are all the bounds we need

CAC again



Book chapter: ON-OFF

If S_i are ON-OFF

$$G_{ON} = \lambda \qquad \overline{T_{ON}} = t_{ON}$$

$$G_{OFF} = 0 \qquad \overline{T_{OFF}} = t_{OFF}$$

Aggregate traffic avg traffic generated by the N source is $G = \sum_{i=1}^{N} G_i$

$$\overline{G_i} = G_{ON} \cdot \underbrace{\frac{t_{ON}}{t_{ON} + t_{OFF}}}_{P_{ON}} \qquad G = N \cdot \overline{G} = N\lambda \cdot \frac{t_{ON}}{t_{ON} + t_{OF}} < C$$

necessary condition for N sources to be admissible

If $B = 0 \rightarrow$ max N we can admit while guaranteeing a cong prob. $\gamma \leq \varepsilon$

Can be computed as follows:

We have congestion whenever

- the aggregate traffic generated by active sources exceeds C

$$G(t) = \sum_{i=1}^{N} \underbrace{G_{ON}}_{=\lambda} \cdot \chi\{\text{source } i \text{ is ON at time } t\}$$

For a generic time t $P[\text{source } j \text{ is active at } t] = P_{ON}$

$$= \frac{t_{ON}}{t_{ON} + t_{OFF}}$$

$A(t) \triangleq$ # of active sources at t

$$= \sum_{i=1}^{N} \chi\{\text{source } "i" \text{ is active @ } t\}$$

$\chi\{\cdot\} \triangleq \chi_i \in \{0, 1\}$

$$A(t) \sim B(N, P_{ON})$$

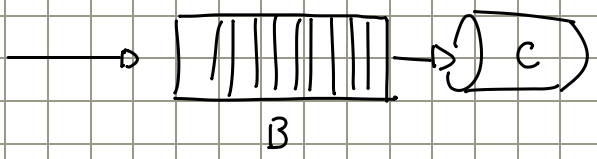
↑ Bernoulli

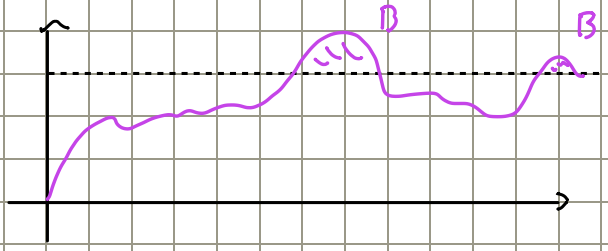
$$P[G(t) > c] = P[\lambda A(t) > c] = P\left(A(t) > \frac{c}{\lambda}\right) \overset{\text{if } \frac{c}{\lambda} < N}{=} \sum_{k=\lceil c/\lambda \rceil}^{N} \binom{N}{k} P_{ON}^{k} (1 - P_{ON})^{N-k}$$

Easy part, now we add the buffer

We can extend the bound to a general case

We have a buffer





Cong condition ?

$P[q(t) = B, \; G(t) > c]$; $P[q(t) > \tilde{B}]$

$$P[q(t) > B] \leq \varepsilon$$